

CNS

By Dr.Riffat Ibrahim Riffat

Local Anesthetics

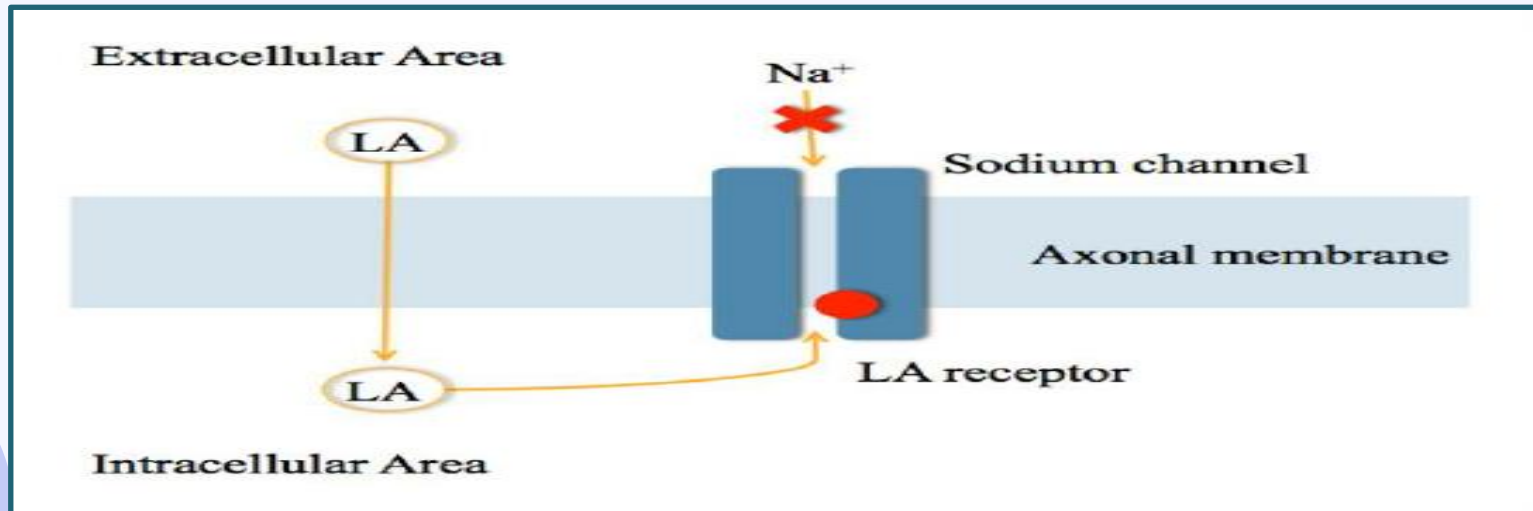


Introduction

local anesthesia is intended to produce Reversible loss of sensation in small part of the body as these used in dentistry and dermatology. Not including the brain.

- Notes: Small “ Type A delta “ and unmyelinated nerve fibers “ Type C” for pain are most sensitive for local anesthetics than the large fibers (motor).

- **MOA:**
- 1- Block voltage gate sodium channel within the nerve fibers so decrease nerve conduction.
- 2- The non ionized form (lipid soluble) cross the lipid membrane.
- 3- Once inside, It is converted into ionized form (active) which bind to the inner vestibule of the sodium channel



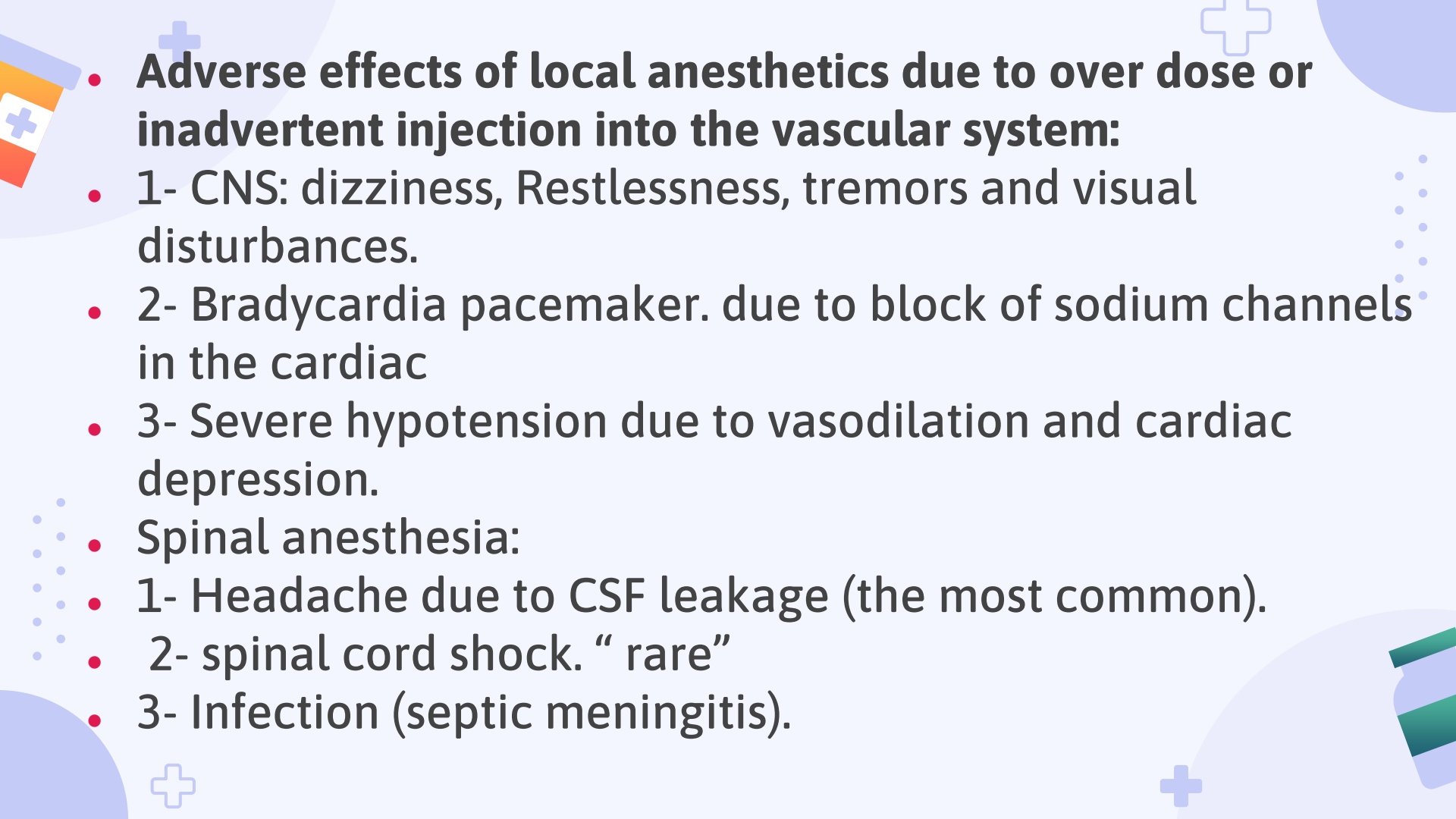


• Local anesthetic drugs:

- Esters: Cocaine, Procaine, Benzocaine.
 - Have short duration and metabolized by plasma and tissues esterase.
 - 1 Procaine: Prototype (The first synthetic agent). No Longer used.
 - 2 Cocaine: Natural source and rarely used.
 - 3 Benzocaine: - It is used for surface (topical) anesthesia only as creams and oral gel for oral ulcers.
 - The topical used of higher concentration of benzocaine (teething gel) may cause methemoglobinemia, this side effect is most common in children less than 2 years.
- 
- 
- 

- **Local anesthetic drugs:**

- **Amides:** (topical local anesthetics for the eyes). **Lidocaine, Mepivacaine, Bupivacaine and Ropivacaine.**
 - Have long duration and metabolized by the liver.
 - Lidocaine (lignocaine): Prototype.
 - Widely used local anesthetics.
 - Used also by IV for ventricular arrhythmia. “ Class Ib”.
 - Used topically as eye drops.
 - All LAs are Vasodilators except cocaine so added to as epinephrine a vasoconstrictor to reduce bleeding , increase duration of anesthesia and decrease amount of the drug from reaching systemic circulation.
- 
- 
- 
- 

- 
- **Adverse effects of local anesthetics due to over dose or inadvertent injection into the vascular system:**
 - 1- CNS: dizziness, Restlessness, tremors and visual disturbances.
 - 2- Bradycardia pacemaker. due to block of sodium channels in the cardiac
 - 3- Severe hypotension due to vasodilation and cardiac depression.
 - Spinal anesthesia:
 - 1- Headache due to CSF leakage (the most common).
 - 2- spinal cord shock. “rare”
 - 3- Infection (septic meningitis).

Cases

- A 32-year-old woman requests an epidural to ease labor pains. She reports that she has an allergic reaction to Novocain(procaine) at the dentist's office. Which of the following local anesthetics would be appropriate for use in an epidural for this patient?
- A. Chloroprocaine.
- B. Mepivacaine.
- C. Ropivacaine.
- D. Tetracaine.

Answers

- Correct answer= C.- Procaine is an ester local anesthetic. Since this patient has an allergy to procaine, other ester anesthetics (Chloroprocaine and tetracaine) should not be used. - Mepivacaine an amide local anesthetics, should not be used due to potential for increased toxicity to the neonate.- Ropivacaine is an amide anesthetic

Cases

- Which of the following is correct regarding local anesthetics?
- A. They affect only small , unmyelinated nerve fibers.
- B. They have either a lipophilic or a hydrophilic group.
- C. They have either an amide or an ester linkage.
- D. They are unaffected by PH of the tissue and pKa of the drug.
- E. In their ionized form , they interact with the protein receptor of calcium channels.

Answers

- Correct answer= C. - The small , unmyelinated nerve fibers that conduct impulses for pain, temperature and autonomic activity are most sensitive to the action of local anesthetics, but other nerve fibers are affected also.- Local anesthetics have a lipophilic group (aromatic ring) , joined by either an amide or ester linkage to a carbon chain that in turn is joined to amine hydrophilic group. - Onset and duration of local anesthetics are affected by both PH of the tissue and PKa of the drug. - Local anesthetics work by blocking sodium ion channels.

Cases

- A 36-year-old man I said to have an endoscopy, which of the following agents can be used as the sole anesthetic for this procedure
 - A. Ketamine
 - B. Propofol
 - C. Midazolam
 - D. sevofulrane

Answers

- Correct answer = D. Midazolam is used as a sole agent for surgical and diagnostic procedures that do not require analgesia (endoscopy, cardiac catheterization).